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The University of Melbourne

Semester 2 Examination —November, 2011

Department of Mathematics and Statistics

MAST30027 Modern Applied Statistics

Exam duration: three hours
Reading time: fifteen minutes
This paper has 8 pages.

Authorised materials:
Hand-held electronic calculators may be used.

Instructions to invigilators:
A 16-page script book shall be supplied to each student.

Instructions to students:
There are 11 questions. All questions may be attempted.
The number of marks for each question is indicated after the question.
The total number of marks available is 100.

This paper may be released to the Library

1. Consider a random sample $X_1, \cdots, X_n$ from a distribution that has the following *pdf*

$$f(x|\theta) = \frac{4}{\theta}(x+2)^3 e^{-\frac{(x+2)^4}{\theta}}, \quad x > -2; \ \theta > 0.$$

Note the following results can be quoted without proof:

(i) $(X_i+2)^4 \stackrel{d}{=} \text{EXP}(\theta), \quad i = 1, \cdots, n.$

(ii) The mean, variance and *mgf* (moment generating function) for exponential distribution EXP(θ) are θ, θ^2 and $(1-\theta t)^{-1}$ with $t < \theta^{-1}$, respectively.

(iii) The minimum variance (i.e. Cramer-Rao lower bound) of an unbiased estimator T of $\tau(\theta)$ is given by the information inequality

$$\text{Var}(T) \geq \frac{[\tau'(\theta)]^2}{-nE\left[\frac{d^2 \ln f(X|\theta)}{d\theta^2}\right]}.$$

Now complete the following tasks:

(a) Specify the joint *pdf* of $X_1, \cdots, X_n$.

(b) Find the score function $U(\theta) = \frac{d \ln L(\theta)}{d\theta}$ and the MLE of θ based on the sample.

(c) Show that the MLE of θ, denoted as $\hat{\theta}$, is unbiased for θ.

(d) Show that the MLE $\hat{\theta}$ is a UMVUE (uniformly minimum variance unbiased estimator).

[2 + 2 + 1 + 4 = 9 marks]

2. The following results can be used for this question:

(i) **Lehmann-Scheffe Theorem**: Suppose $\mathbf{S} = (S_1, \cdots, S_k)$ is jointly sufficient and complete for $\boldsymbol{\theta}$. If $T^* = t^*(\mathbf{S})$ is an unbiased estimator of $\tau(\boldsymbol{\theta})$, then T^* is a UMVUE of $\tau(\boldsymbol{\theta})$.

(ii) A *pdf* is a member of the **regular exponential class (REC)** if it has the following form

$$f(x|\boldsymbol{\theta}) = c(\boldsymbol{\theta})h(x)\exp\left(\sum_{j=1}^{q}\eta_j(\boldsymbol{\theta})t_j(x)\right), \quad x \in A$$

and 0 elsewhere, where $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_q)$ with q unknown components has a rectangular parameter space; and provided that the following conditions are satisfied:

1° $A = \{x : f(x|\boldsymbol{\theta}) > 0\}$ does not depend on $\boldsymbol{\theta}$;

2° $\eta_j(\boldsymbol{\theta})$'s are nontrivial, functionally independent and continuous functions of $\boldsymbol{\theta}$;

3° $t_j(x)$'s are nontrivial and their derivatives are linearly independent.

(iii) Binomial distribution BIN(m, θ) has mean $m\theta$, variance $m\theta(1-\theta)$ and *mgf* $[\theta(e^t-1)+1]^m$.

Consider a random sample $X_1, \cdots, X_n$ from a BIN(m, θ) distribution that has the *pdf*

$$f(x|\theta) = \binom{m}{x}\theta^x(1-\theta)^{m-x}, \quad x = 0, 1, \cdots, m; \quad 0 < \theta < 1; \quad \text{and } m \text{ is known.}$$

(a) Show why BIN(m, θ) belongs to REC.

(b) Find a complete and sufficient statistic for θ based on the random sample $X_1, \cdots, X_n$.

(c) Define $\phi = \theta(1-\theta)$. Show that $\hat{\phi} = \frac{\sum_{i=1}^n X_i[mn - \sum_{i=1}^n X_i]}{mn(mn-1)}$ is an unbiased estimator of ϕ.

(d) Is $\hat{\phi}$ a UMVUE of ϕ? Why or why not?

[2 + 1 + 4 + 2 = 9 marks]

3. Consider a GLM (generalised linear model) for a random sample $Y_1, \cdots, Y_n$ which is to be observed from a Poisson distribution. Here each Y_i has the *pdf*

$$f(y_i|\mu_i) = \frac{\mu_i^{y_i}}{y_i!}e^{-\mu_i}, \quad y_i = 0, 1, 2, \cdots; \ \mu_i > 0; \quad i = 1, \cdots, n;$$

where $\mu_i = E(Y_i)$. Suppose the systematic component of the GLM is $\eta_i = \alpha + \beta x_i$, with (α, β) being the parameters of interest and x_i's being the covariate observations; and the link is $\eta_i = \ln(\mu_i)$, $i = 1, \cdots, n$.

(a) Find the log-likelihood function $\ell(\alpha, \beta) = \ln L(\alpha, \beta)$.

(b) Find $\frac{\partial \ln L}{\partial \alpha}$ and $\frac{\partial \ln L}{\partial \beta}$ which specify the score function $U(\alpha, \beta)$.

(c) Find $\frac{\partial^2 \ln L}{\partial \alpha^2}$, $\frac{\partial^2 \ln L}{\partial \beta^2}$ and $\frac{\partial^2 \ln L}{\partial \alpha \partial \beta}$. Then list the Fisher information matrix $I(\alpha, \beta)$.

(d) Given that $(\alpha^{(j)}, \beta^{(j)})$ is the current estimate in the process of finding the MLE of (α, β), we can update it to $(\alpha^{(j+1)}, \beta^{(j+1)})$ by the Newton-Raphson or Fisher scoring procedure. Specify the formula used in this update.

[2 + 2 + 4 + 2 = 10 marks]

4. A size-n random sample $(Y_1, X_1), \cdots, (Y_n, X_n)$ satisfy the model $Y_i = X_i + \varepsilon_i$, $i = 1, \cdots, n$, where each X_i has the pdf $f(x_i|\lambda) = \lambda e^{-\lambda x_i}$, $x_i > 0$, $\lambda > 0$; each ε_i has the pdf $g(\varepsilon_i) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}\varepsilon_i^2}$, $-\infty < \varepsilon_i < \infty$; and all X_i's and ε_i's are mutually independent.

Suppose we have observed $y_1, \cdots, y_n$ for $Y_1, \cdots, Y_n$ only. We want to use an EM algorithm to find the MLE (maximum likelihood estimate) of λ.

(a) We may treat $(Y_1, X_1), \cdots, (Y_n, X_n)$ as the complete data. Find the joint pdf for each (Y_i, X_i) pair. Then find the complete-data log-likelihood function $\ell(\lambda) = \ln L(\lambda)$.

(b) Show that the conditional pdf of X_i given $Y_i = y_i$ is

$$f(x_i|y_i; \lambda) = \frac{\phi(x_i - (y_i - \lambda))}{\Phi(y_i - \lambda)}, \quad x_i > 0$$

where $\phi(t) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}t^2}$ and $\Phi(s) = \int_{-\infty}^{s} \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}t^2}dt$ are the standard normal pdf and cdf respectively.

(c) In the E-step, we calculate

$$Q(\lambda, \lambda^{(k)}) = E[\ln L(\lambda) \mid Y_1 = y_1, \cdots, Y_n = y_n; \lambda^{(k)}]$$

where $\lambda^{(k)}$ is the current estimate of λ. Show that

$$\begin{aligned} Q(\lambda, \lambda^{(k)}) &= -n \ln \sqrt{2\pi} + n \ln \lambda - \frac{1}{2}\sum_{i=1}^{n} y_i^2 + \sum_{i=1}^{n}(y_i - \lambda)\left(y_i - \lambda^{(k)} + \frac{\phi(y_i - \lambda^{(k)})}{\Phi(y_i - \lambda^{(k)})}\right) \\ &\quad -\frac{1}{2}\sum_{i=1}^{n}\left(1 + (y_i - \lambda^{(k)})^2 + \frac{(y_i - \lambda^{(k)})\phi(y_i - \lambda^{(k)})}{\Phi(y_i - \lambda^{(k)})}\right). \end{aligned}$$

(*Note*: The following results can be quoted without proof:
$E(X_i|Y_i = y_i; \lambda) = y_i - \lambda + \frac{\phi(y_i - \lambda)}{\Phi(y_i - \lambda)}$ and $E(X_i^2|Y_i = y_i; \lambda) = 1 + (y_i - \lambda)^2 + \frac{(y_i - \lambda)\phi(y_i - \lambda)}{\Phi(y_i - \lambda)}$.)

(d) In the M-step, we maximise $Q(\lambda, \lambda^{(k)})$ with respect to λ to find an update $\lambda^{(k+1)}$ from $\lambda^{(k)}$. Show that

$$\lambda^{(k+1)} = \frac{n}{\sum_{i=1}^{n}\left(y_i - \lambda^{(k)} + \frac{\phi(y_i - \lambda^{(k)})}{\Phi(y_i - \lambda^{(k)})}\right)}.$$

[3 + 3 + 3 + 1 = 10 marks]

5. Let $X_1, \cdots, X_n$ be a random sample from negative binomial distribution with $X_i \stackrel{d}{=} \text{NB}(r_i, \theta)$ and r_i being given. Thus the *pdf* of X_i is

$$f(x_i|\theta) = \binom{x_i - 1}{r_i - 1}\theta^{r_i}(1-\theta)^{x_i - r_i}, \quad x_i = r_i, r_i + 1, \cdots; \quad i = 1, \cdots, n.$$

Suppose θ has a BETA(a, b) prior distribution with the *prior pdf* being

$$p(\theta|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}; \quad a > 0, \ b > 0 \text{ being given}; \quad \text{also } E(\theta) = \tfrac{a}{a+b}.$$

(a) Find the posterior pdf of θ given $X_1, \cdots, X_n$.

(b) Find the Bayes estimator of θ when the squared loss function is used.

[5 + 3 = 8 marks]

6. A random number X is to be generated by the following acceptance-rejection (A-R) sampling algorithm:

1° Generate $U = u$ from Unif$(0, 1)$ and $V = v$ from Unif$(0, 1)$ independently.

2° If $u + v > 1$, deliver $X = \frac{1}{u^3}$; otherwise, go to 1°.

The *cdf* (cumulative distribution function) of X can be found using the following result:

$$P(X \le x) = P(\frac{1}{U^3} \le x \mid U + V > 1), \quad \text{for a given } x.$$

(a) Show that the pdf of X is $f(x) = \frac{2}{3}x^{-5/3}, \quad x > 1.$

(b) Let N be the number of times steps 1° and 2° need be repeated before an X value is generated. What is the distribution of N and what is the mean $E(N)$?

[5 + 3 = 8 marks]

7. Consider a random sample $X_1, \cdots, X_n$ from GAMMA$(\theta, 4)$ distribution which has the pdf

$$f(x|\theta) = \frac{1}{6\theta^4}x^3 e^{-\frac{x}{\theta}}; \quad x > 0, \ \theta > 0.$$

It can be shown that θ is a scale parameter and $\hat{\theta} = \frac{1}{4}\bar{X}_n = \frac{1}{4n}\sum_{i=1}^n X_i$ is the MLE of θ. It can also be shown that $Q = \frac{2n\bar{X}_n}{\theta} \stackrel{d}{=} \chi^2(8n)$ and Q is a pivotal quantity.

(a) Explain what is called a pivotal quantity.

(b) Derive a 95% C.I. (confidence interval) formula for θ based on Q.

(c) Based on a sample of 50 observations from GAMMA$(\theta, 4)$, 1999 nonparametric bootstrap samples are generated to estimate se$(\hat{\theta})$ and construct C.I. for θ. The following R output is obtained where x stores the 50 generated GAMMA$(\theta, 4)$ observations:

```
> library(boot)
> scale.bt=function(x,i){scale=mean(x[i])/4
           n=length(x)
          v.scale=(n-1)*var(x[i])/(16*n^2)
          c(scale,v.scale)}
> set.seed(12345)
> bootQ7=boot(data=x,statistic=scale.bt,R=1999)
> bootQ7
ORDINARY NONPARAMETRIC BOOTSTRAP

Call:
boot(data = x, statistic = scale.bt, R = 1999)

Bootstrap Statistics :
     original        bias    std. error
t1* 6.0065836 -0.0006930831  0.46658707
t2* 0.2156650 -0.0044079634  0.03950498
```

```
> boot.ci(bootQ7)

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1999 bootstrap replicates

CALL :
boot.ci(boot.out = bootQ7)

Intervals :
Level      Normal              Basic              Studentized
95%   ( 5.093,  6.922 )   ( 5.064,  6.914 )   ( 5.087,  7.026 )

Level     Percentile            BCa
95%   ( 5.099,  6.949 )   ( 5.124,  6.980 )
Calculations and Intervals on Original Scale

> qchisq(0.975,400); qchisq(0.025,400)

[1] 457.3055
[1] 346.4818
```

i. From the R output find the value of $\hat{\theta}$, the bootstrap estimate of se$(\hat{\theta})$, and the 95% Studentized C.I. of θ. Explain how the 95% Studentized C.I. of θ is calculated.

ii. Calculate the 95% C.I. for θ using the formula obtained in (b) and the χ^2 percentile values in the R output.

[1 + 3 + 4 + 1 = 9 marks]

8. **Neyman-Pearson Lemma:** Suppose random variables $X_1, \cdots, X_n$ have the joint pdf $f(\mathbf{x}_n|\theta)$ with $\mathbf{x}_n = (x_1, \cdots, x_n)^T$. Let $\lambda(\mathbf{x}_n) = \dfrac{f(\mathbf{x}_n|\theta_1)}{f(\mathbf{x}_n|\theta_0)}$ be the likelihood ratio test statistic, and $C^* = \{\mathbf{x}_n : \lambda(\mathbf{x}_n) \geq k\}$ be a critical region for testing

$$H_0\colon \theta = \theta_0 \quad \text{vs.} \quad H_a\colon \theta = \theta_1,$$

where k is such that $P(\mathbf{X}_n \in C^* \mid \theta_0) = \alpha$. Then C^* is the most powerful critical region of size α for the test. Namely, for any critical region C such that $P(\mathbf{X}_n \in C|\theta_0) \leq \alpha$,

$$\pi_{C^*}(\theta_1) = P(\mathbf{X}_n \in C^* \mid \theta_1) \geq P(\mathbf{X}_n \in C \mid \theta_1) = \pi_C(\theta_1).$$

Question: Consider a random sample $X_1, \cdots, X_n$ from a Weibull distribution WEI$(\theta, 3)$ which has the *pdf*

$$f(x|\theta) = \frac{3}{\theta^3} x^2 e^{-\left(\frac{x}{\theta}\right)^3}; \quad x > 0, \theta > 0.$$

It can be shown that $X_i^3 \stackrel{d}{=} \text{EXP}(\theta^3)$, $i = 1, \cdots, n$; and $\hat{\theta} = \left(\frac{1}{n}\sum_{i=1}^n X_i^3\right)^{\frac{1}{3}}$ is the MLE of θ.

(a) Determine an MP (most powerful) critical region for testing $H_0\colon \theta = \theta_0$ vs. $H_a\colon \theta = \theta_1$, where $\theta_1 > \theta_0$, at significance level α. Simplify the critical region as much as you can.

(b) Explain whether the critical region found in (a) is UMP (uniformly most powerful) of size α for testing $H_0\colon \theta \leq \theta_0$ vs. $H_a\colon \theta > \theta_0$.

(c) Derive a GLR (generalised likelihood ratio) test for testing $H_0\colon \theta = \theta_0$ vs. $H_a\colon \theta \neq \theta_0$. Simplify the derived critical region as much as you can.

[3 + 3 + 3 = 9 marks]

9. Thirty recreational basketball players were asked to shoot two free throws. Data on whether they made or missed their shots are shown in the table.

	Missed the 2nd attempt	Made the 2nd attempt
Missed the 1st attempt	8	10
Made the 1st attempt	2	10

The question of interest is whether the probability of making a shot on the first attempt is different from the probability of making a shot on the second attempt. This can be answered by using McNemar's method to test H_0: $p_1 = p_2$ vs. H_a: $p_1 \neq p_2$, where p_1 is an individual's probability of making a shot on the first attempt and p_2 is that on the second attempt.

(a) Let Y_{12} denote the top-right observation in the table. What is the distribution of Y_{12} when H_0 is true? Find the exact p-value of the test using McNemar's binomial test statistic.

(b) The following is the R output from the McNemar's χ^2 test. Explain how the test statistic value $\chi^2 = 4.0833$ is obtained.

```
> x=matrix(c(8,2,10,10),2,2)
> mcnemar.test(x, correct=T)

        McNemar's Chi-squared test with continuity correction

data:  x
McNemar's chi-squared = 4.0833, df = 1, p-value = 0.04331
```

[3 + 3 = 6 marks]

10. The following example comes from Maindonald and Braum (2003, p.199). Thirty patients were given an anesthetic agent that was maintained at a pre-determined (alveolar) concentration for 15 minutes before making an incision. It was then noted whether the patient moved, i.e., jerked or twisted. The interest is in estimating how the probability of jerking or twisting varies with increasing concentration of the anesthetic agent.

The following table lists the data, where y is the number of patients who moved among the n patients whose concentration level of the received anesthetic agent was maintained at x.

y	6	4	2	2	0	0
n	7	5	6	6	4	2
x	0.8	1.0	1.2	1.4	1.6	2.5

A logistic model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$, where θ is the probability of a patient moving (i.e., jerking or twisting), is fitted to the data. The relevant results are provided in the following R output.

```
> anes.2 <- glm(y/n ~ x,family=binomial,weight = n,data=anes.dat)
> anova(anes.2, test="Chi")

Analysis of Deviance Table

Model: binomial, link: logit

Response: y/n

Terms added sequentially (first to last)


     Df Deviance Resid. Df Resid. Dev P(>|Chi|)
NULL                     5    15.4334
x     1   13.701         4     1.7321 0.0002143

> 1-pchisq(1.7321,4)

[1] 0.7848787

> summary(anes.2)$coef

             Estimate Std. Error   z value     Pr(>|z|)
(Intercept)  6.468675   2.418537  2.674624 0.007481321
x           -5.566762   2.043649 -2.723932 0.006450977
```

```
> summary(anes.2)$cov.unscaled

            (Intercept)         x
(Intercept)    5.849319 -4.848448
x             -4.848448  4.176502
```

(a) Use both Wald test and the (generalised) likelihood ratio test to assess the significance of concentration level x on affecting the probability of moving based on model `anes.2`. Report the test statistic value and the p-value for each test.

(b) Does the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ provide an adequate fit to the data? Why or why not?

(c) Estimate the odds ratio of a patient moving when the concentration level x decreases by 0.1 unit. Also find an approximate 95% C.I. for this odds ratio.

(d) Estimate the probability of a patient moving when $x = 1.8$. Also find an approximate 95% C.I. for this probability when $x = 1.8$.

[$2 + 2 + 3 + 4 = 11$ marks]

11. The data given in the following $2 \times 2 \times 3$ table was used to study the effect of passive smoking on lung cancer. The table summarizes the results of case-control studies from three countries for nonsmoking women married to smokers. (Source: Blot and Fraumeni, J. Nat. Cancer Inst., 77:993-1000 (1986) and Agresti (1996).)

Country	Spouse Smoked	Cases	Controls
Japan	No	21	82
	Yes	73	188
UK	No	5	16
	Yes	19	38
USA	No	71	249
	Yes	137	363

(a) A Mantel-Haenszel test is performed returning the following R output. Explain what this test is testing for and what the conclusion is at the significance level 0.05.

```
> ex5=array(0,c(2,2,3))
> ex5[,,1]=c(21,73,82,188)
> ex5[,,2]=c(5,19,16,38)
> ex5[,,3]=c(71,137,249,363)
> mantelhaen.test(ex5,correct=T)

        Mantel-Haenszel chi-squared test with continuity correction

data:  ex5
Mantel-Haenszel X-squared = 5.138, df = 1, p-value = 0.02341
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 0.5489352 0.9491681
sample estimates:
common odds ratio
        0.7218253
```

(b) The data in the table are rearranged so that a log-linear model `mod1` can be fitted, with the results being given in the following R output. Give the mathematical formula (of the form $\ln(\lambda) = \cdots$) for model `mod1`.

```
> pasSmoking.dat=data.frame(freq=c(21,73,5,19,71,137,82,188,16,38,249,363))
> pasSmoking.dat$Cnt=factor(rep(c("Japan","UK", "USA"), times=2, each=2))
> pasSmoking.dat$Smo=factor(rep(c("No","Yes"), times=6))
> pasSmoking.dat$Can=factor(rep(c("Case","Control"), each=6))
> pasSmoking.dat
```

```
   freq   Cnt Smo     Can
1    21 Japan  No    Case
2    73 Japan Yes    Case
3     5    UK  No    Case
4    19    UK Yes    Case
5    71   USA  No    Case
6   137   USA Yes    Case
7    82 Japan  No Control
8   188 Japan Yes Control
9    16    UK  No Control
10   38    UK Yes Control
11  249   USA  No Control
12  363   USA Yes Control
```

```
> mod1=glm(freq~Cnt+Smo+Can+Cnt:Smo+Cnt:Can+Smo:Can, family=poisson, data=pasSmoking.dat)
> anova(mod1, test="Chisq")
Analysis of Deviance Table

Model: poisson, link: log

Response: freq

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev P(>|Chi|)
NULL                       11    1168.85
Cnt      2   726.43         9     442.42 < 2.2e-16
Smo      1   112.52         8     329.90 < 2.2e-16
Can      1   307.56         7      22.34 < 2.2e-16
Cnt:Smo  2    15.50         5       6.84 0.0004316
Cnt:Can  2     1.05         3       5.80 0.5919109
Smo:Can  1     5.56         2       0.24 0.0184215
> 1-pchisq(0.24,2)
[1] 0.8869204
> 1-pchisq(5.80,3)
[1] 0.1217566
```

(c) What assumption is made for the Mantel-Haenszel test in (a) to be valid, in regard to the association between the women's lung cancer and passive smoking from their spouses? Test this assumption using the R output in (b) and draw the conclusion at the significance level 0.05.

(d) Test based on `mod1` the significance of the interaction effect `Smo:Can`, eliminating the effects of all other terms in the model. Comment on the implication of your result and how that is related to the result of (a).

(e) Test the adequacy of model `Cnt+Smo+Can+Cnt:Smo+Cnt:Can`, at significance level 0.05, using the R output in (b). Comment on the implication of your result and how that is related to the results of (a) and (d).

$[2+2+3+2+2=11$ marks$]$

Total marks = 100

END OF EXAMINATION